

Worksheet 5: Externalities

This worksheet is graded for completion since it is due the same week as our last exam. However, please do attempt every problem and do not just write something down to get points as that won't suffice. This will be good practice for Exam 2.

Scenario A

Consider the market for a specific fruit product (GrapplesTM) where the demand is given by:

$$Q_D = 10,000 - 400P$$

The inverse supply (i.e. "Private Marginal Cost") is:

$$PMC = 5 + 0.0075Q$$

The production of Grapples uses large amounts of fertilizer which results in runoff and damage to the water in adjacent rivers and streams. The marginal cost of this pollution is given by the Marginal External Cost equation:

$$MEC = 0.0025Q$$

Grapple producers are not liable for these harms and so ignore them in their production decisions.

1. What is the inverse demand equation (AKA "private marginal benefit")?
2. Find the competitive equilibrium Quantity of Grapples.
3. Find the competitive equilibrium Price of Grapples.
4. Find the Consumer Surplus in equilibrium.
5. Find the Producer Surplus in equilibrium.
6. What is the total external cost in equilibrium? *Hint: The total external cost is the area below the marginal external cost curve.*
7. When accounting for the external cost, what is the efficient quantity of output in the market?
8. What is the Deadweight Loss created by the externality?
9. In order to reduce pollution, the government decides to tax Grapple growers. The tax is set as a per-unit (per Grapple) amount. What tax per Grapple will result in the efficient outcome?
10. What is the deadweight loss in the market after the tax is instituted?
11. Suppose that the government doesn't impose a tax on the market. Instead, the government allows a monopoly to form. The monopoly has the same private marginal cost as the market supply given earlier. How many Grapples would a monopoly produce?
12. Is the monopoly producing the efficient level of output? (multiple choice)
 - A. No, the monopoly produces more than the efficient quantity
 - B. Yes, the output of the monopoly is the same as the efficient quantity
 - C. Yes, because the monopoly will take into account the external cost when choosing the quantity of Grapples to sell
 - D. No, the monopoly produces less than the efficient quantity

Scenario B

The inverse demand (AKA “private marginal benefit”) for a vaccine against a specific disease is given by

$$P = 13 - 0.0005Q$$

where Q is the number of doses/shots given. The supply (AKA “private marginal cost”) is horizontal at $P = 8$. In other words, the marginal and average cost of manufacturing the vaccine is constant at \$8.

The vaccine also creates a positive external benefit in reducing the spread of the disease to others. The Marginal External Benefit is given by:

$$MEB = 7 - 0.00025Q$$

13. Suppose there is a competitive market for the vaccines. What is the competitive equilibrium quantity?
14. What is the competitive market equilibrium price for vaccines?
15. When accounting for the external benefit of the vaccine, what is the efficient quantity of vaccines?
16. What is the deadweight loss created by the positive externality?
17. Suppose the government decides to encourage the efficient quantity of vaccines by providing a subsidy. The subsidy is set at a specified dollar amount per dosage purchased. What subsidy (\$) will lead to the efficient quantity of doses being administered?
18. How much will the government need to spend on the subsidy in total? (i.e. “Government Expenditure”)
19. Is the benefit of the subsidy found above greater than the government’s cost of the subsidy? (Multiple Choice)
 - A. No because the consumer surplus and producer surplus increase by an amount less than the government’s expenditure on the subsidy
 - B. Yes because the government’s expenditure is less than the sum of the added consumer and producer surplus and added external benefit
 - C. Yes because the subsidy will increase producer and consumer surplus by an amount more than the subsidy
 - D. Yes because the government’s expenditure is less than the reduction in deadweight loss
 - E. Yes because the sum of consumer and producer surplus is more than the total external benefit